

SUPPORT VECTOR MACHINE (SVM) :-

- SVM is a supervised classification algorithm.
- Instead of separating classes, it asks "which boundary separate them with the maximum safety gap (margin)?"
- Effective in high-dimensional spaces and when the number of features is large.

OPTIMAL HYPERPLANE

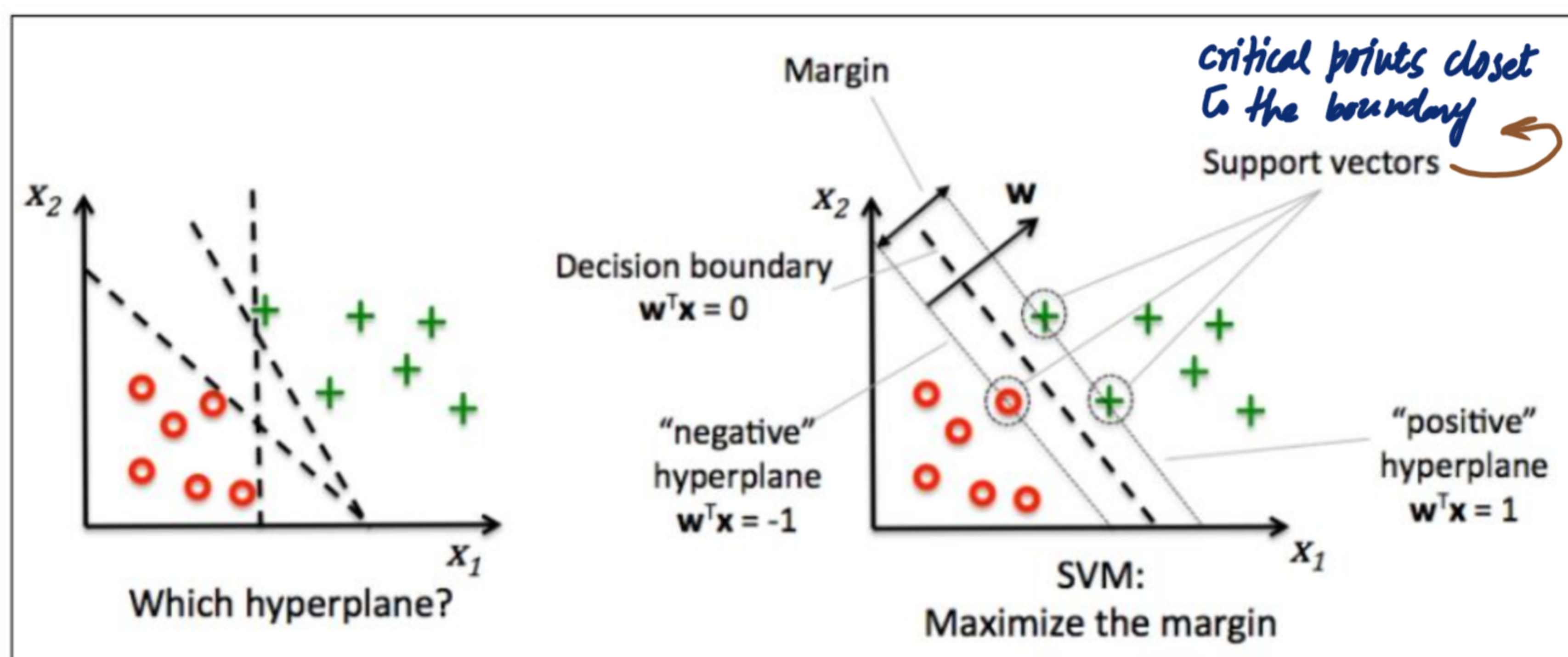
DISTANCE B/W HYPERPLANE AND NEAREST DATA POINTS

MATHEMATICAL FORMULATION :-

- LINEAR SVM $\rightarrow w^T x + b = 0$
- NON-LINEAR SVM $\rightarrow k(x_i, x_j) = \phi(x)^T \phi(x_j)$

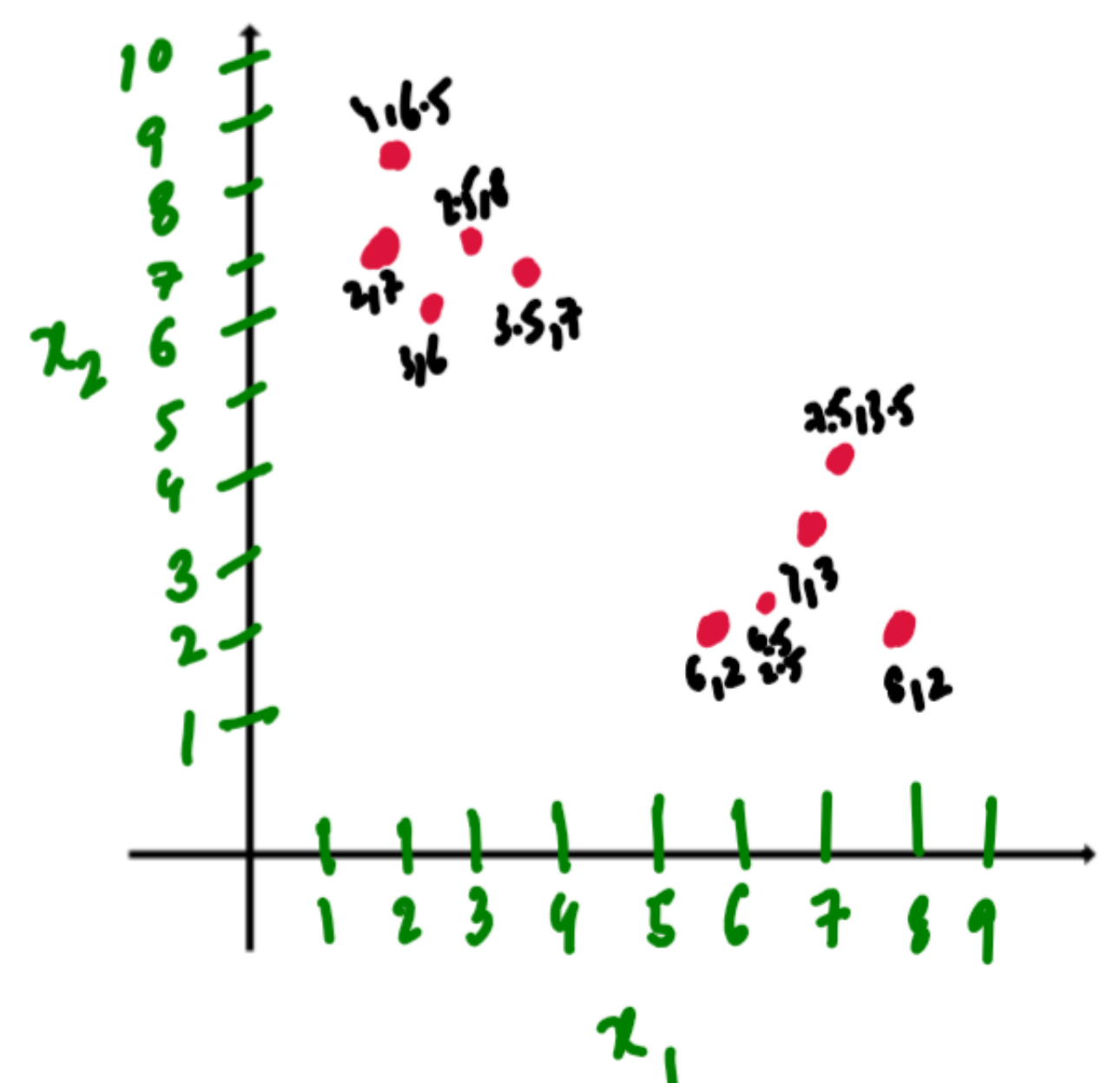
w = weight vector ; b = bias

x = input vector



EXAMPLE: A factory evaluates steel sheets based on x_1 thickness (mm) and x_2 Surface Roughness and decides to accept or reject the sheet

Sheet	x_1	x_2	Class
1	6	2	+1
2	7	3	+1
3	8	2	+1
4	6.5	2.5	+1
5	7.5	3.5	+1
6	2	7	-1
7	3	6	-1
8	2.5	8	-1
9	3.5	7	-1
10	4	6.5	-1



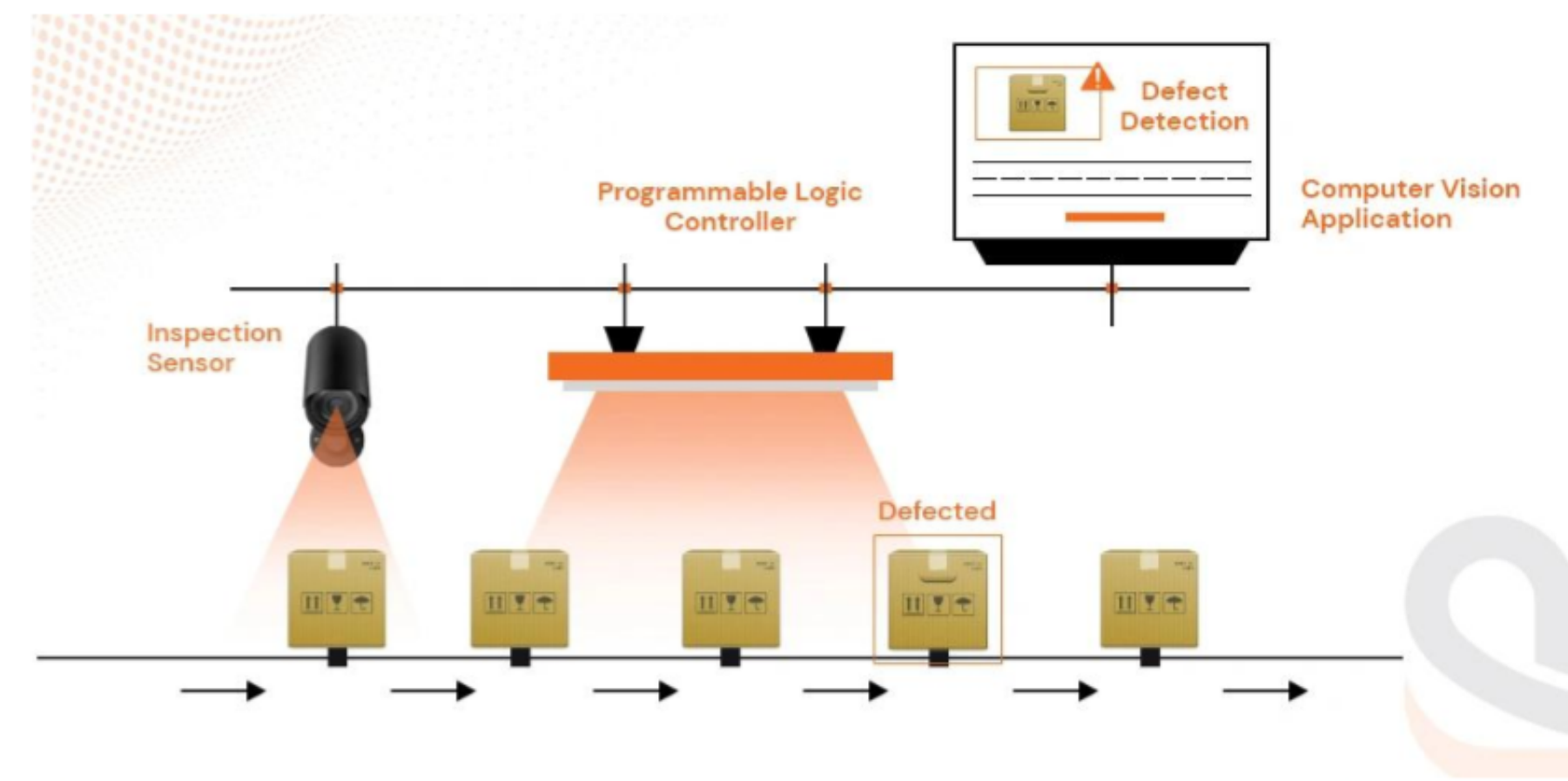
→ STEP #1: UNDERSTAND DATA PATTERN

GOOD SHEETS

- * High thickness
($x_1 \rightarrow \text{large}$)
- ** Lower roughness
($x_2 \rightarrow \text{small}$)

DEFECTIVE SHEETS

- * Lower thickness
($x_1 \rightarrow \text{small}$)
- ** Higher roughness
($x_2 \rightarrow \text{large}$)



STEP #2: IDENTIFY SUPPORT VECTORS

- Representative dataset points

Positive

6, 2

Negative

3.5, 7

STEP #3: CONSTRUCT MARGIN EQUATIONS

$$6w_1 + 2w_2 + b = +1$$

+ve hyperplane

$$3.5w_1 + 7w_2 + b = -1$$

-ve //

Assume $w_1 = 1$

$$\rightarrow 6 + 2w_2 + b = 1 \Rightarrow 2w_2 + b = -5$$

$$\rightarrow 3.5 + 7w_2 + b = -1 \Rightarrow 7w_2 + b = -4.5$$

$$-5w_2 = -7.5$$

$$w_2 = \frac{7.5}{5}$$

Using $w_2 = 7.5/5$

$$\rightarrow 6 + 2\left(\frac{7.5}{5}\right) + b = 1 \Rightarrow 6 + 3 + b = 1$$
$$b = -8$$

STEP #4: FINAL MODEL

$$f(x) = x_1 + \left(\frac{7.5}{5}\right)x_2 - 8$$

STEP #5: DECISION RULE

if $f(x) > 0$ then the sheet will be accepted; else rejected.

Classify following

- If sheet thickness is 4 and roughness is 5, will you accept or reject it?